

# Document Listing V2 - Architecture & Design

This document outlines the comprehensive architecture, design specifications, and implementation details of the Document Listing V2 solution. It covers User Experience (UX), Data Structures, Communication, Automation, and Deployment.

## 1. Project Overview & Use Cases

The **Document Listing V2** is a persistent, theme-aware SharePoint Framework (SPFx) web part designed to display a searchable and filterable catalog of documents from a specified source library. It provides distinct features for categorizing content, requesting access to specific documents, and managing those requests via a separate SharePoint list.

### Key Use Cases

#### 1. Centralized Policy & Procedure Hub (Organizations)

**Scenario:** An organization wants to showcase standard operating procedures (SOPs), HR policies, and compliance documents in a visually appealing, easy-to-navigate format.

- **Need:** Users are overwhelmed by traditional folder structures and cannot find relevant documents quickly.
- **Solution:** The web part categorizes documents by Department (Category) and Doc Type (SubCategory), allowing employees to filter and find exactly what they need in seconds.
- **Value:** improved discoverability and a consistent, branded user experience.

#### 2. Controlled Document Distribution (Compliance/Legal)

**Scenario:** A legal team needs to share sensitive contract templates but must track who downloads them to ensure they are using the latest version.

- **Need:** Prevent uncontrolled proliferation of outdated templates.
- **Solution:** "Request Access" mode is enabled. Users can see the document exists but must click "Request Access" to get the download link.
- **Value:** The system logs every request, providing an audit trail of who requested which document and when.

#### 3. Closed-Loop Feedback System (Knowledge Management)

**Scenario:** A Knowledge Management team wants to not only share best practices but also understand if they are actually helpful to the workforce.

- **Need:** Qualitative data on document utility.
- **Solution:**
  1. User requests a document.
  2. System tracks the download.
  3. After X days, a Power Automate flow sends a follow-up email asking "Was this document helpful?".
  4. A **separate Feedback Web Part** collects this structured feedback.

- **Value:** actionable insights to improve content quality based on real user feedback.

#### 4. Self-Service Template Repository (IT/PMO)

**Scenario:** A Project Management Office (PMO) provides standard project charters and risk logs.

- **Need:** Reduce the burden on PMO staff emailing templates manually.
- **Solution:** Documents are listed in the web part. Users self-serve by searching and downloading. High-value templates can still require a lightweight "Request" to track usage statistics without a heavy approval process.
- **Value:** empowering users while maintaining visibility into asset utilization.

## 2. Architecture Diagram



## 3. Technical Stack

- **Framework:** SharePoint Framework (SPFx)
- **Frontend:** React (Functional Components + Hooks)
- **UI Library:** Fluent UI (Office UI Fabric)
- **Data Access:** SPHttpClient (REST API)
- **Styling:** SCSS Modules with Theme Token support
- **Automation:** Power Automate (Flow)
- **Messaging:** Exchange Online

## 4. UX Layer (SPO > SPFx)

The User Experience is delivered via SharePoint Online (SPO) using the SharePoint Framework (SPFx).

### Key Features

- **Dynamic Catalog:** Fetches documents from a SharePoint Document Library.
- **Multi-Level Filtering:** Side Navigation (Category), Top Tabs (SubCategory), and Search Bar.
- **Access Request Workflow:** Users can request documents; requests are logged in a separate list to prevent duplicates and enable approval flows.
- **Premium UI/UX:** Theme-aware contrast, transparent icons, and responsive layout.

### Visual Requirements

- **Components:** Fluent UI details list, command bar, and panel interactions.
- **Branding:** Header opacity and text color automatically adapt to site theme.
- **Icons:** Standard Fluent file type icons with transparent backgrounds.

## 5. Data Layer & Schema

Data is structured across SharePoint Lists and Libraries to separate content from operational data.

### A. User Guides (Source Document Library)

Role: **Content Storage**.

Permissions: **Read-Only** for general users.

| Internal Name | Type   | Purpose                       |
|---------------|--------|-------------------------------|
| FileLeafRef   | Text   | Filename                      |
| Title         | Text   | Display Title                 |
| Category      | Choice | Primary grouping (Side Nav)   |
| SubCategory   | Choice | Secondary grouping (Top Tabs) |
| Description   | Note   | Details shown in grid         |
| IsPinned      | Yes/No | Top-of-list pinning logic     |

### B. Feedback Tracking (Request Access List)

Role: **Transactional Logging**.

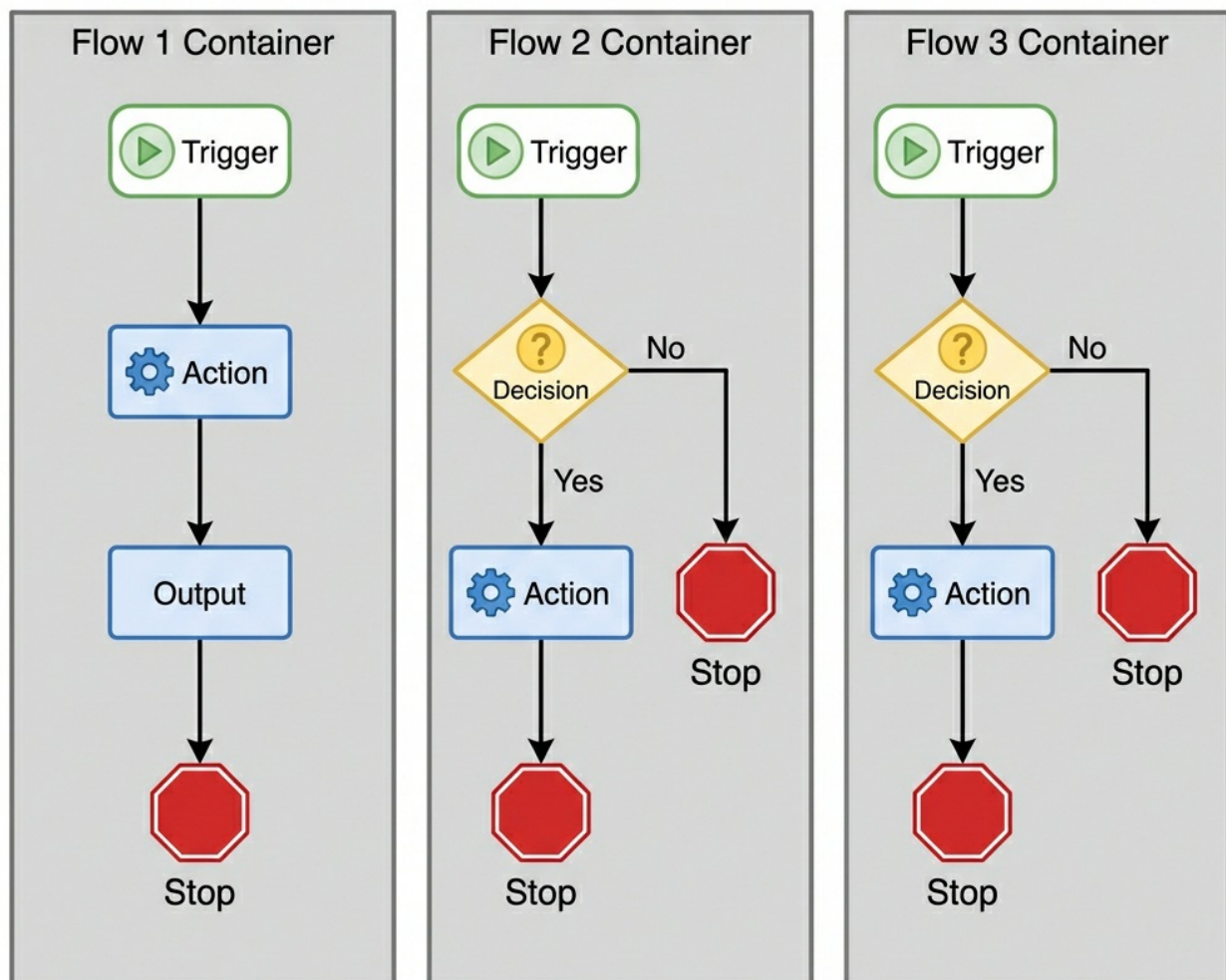
Permissions: **Contribute** (Users create items).

| Internal Name | Type | Purpose                  |
|---------------|------|--------------------------|
| Title         | Text | "Request for [Doc Name]" |
| Email         | Text | Requester's email        |

|                  |          |                            |
|------------------|----------|----------------------------|
| FileID           | Text     | ID of requested document   |
| RequestID        | Text     | Unique GUID                |
| RequestDate      | DateTime | Timestamp of request       |
| DownloadDate     | DateTime | When file was accessed     |
| FeedbackProvided | Yes/No   | Has feedback been given?   |
| FeedbackRequired | Yes/No   | System flag for automation |

## 6. Automation Layer (Power Automate)

### Independent Power Automate Flows



Business logic is decoupled from the web part and handled by Power Automate flows.

#### Flow 1: Send Download Link

- **Trigger:** Item Created in "Request Access List".
- **Action:** Generates a sharing link or retrieves the direct link.
- **Output:** Sends an email to the user with the "Download Document" button.

## Flow 2: Send Feedback Link

- **Trigger:** Scheduled (e.g., 14 days after Download).
- **Logic:** IF Feedback Provided is No -> Send Request.

## Flow 3: Send Feedback Reminder

- **Trigger:** Scheduled (Follow-up to Flow 2).
- **Logic:** IF Feedback Required is Yes AND Time > Threshold -> Send Nudge.

# 7. Prerequisites & Security Model

Successful deployment requires strict adherence to the following configuration models.

## Web Part Configuration

The web part is configured via three primary groups in the Property Pane.

### 1. Display Settings

Controls the visual header and text styling.

| Property              | Type      | Description                                        |
|-----------------------|-----------|----------------------------------------------------|
| Web Part Title        | Text      | Header title (e.g., "Company Policies").           |
| Title Font Size       | Dropdown  | Size of the title text (16px - 32px).              |
| Web Part Description  | Multiline | Subtitle or instructions for users.                |
| Description Font Size | Dropdown  | Size of the description text (12px - 18px).        |
| Header Opacity        | Slider    | Transparency of the header background (0.0 - 1.0). |

### 2. Document Library Settings

Configures the source of the documents.

| Property                | Type        | Description                                    |
|-------------------------|-------------|------------------------------------------------|
| Select Site             | Site Picker | The SharePoint site hosting the documents.     |
| Source Document Library | Dropdown    | The specific library to display.               |
| Category Field          | Dropdown    | Column for Side Navigation grouping.           |
| Sub-Category Field      | Dropdown    | Column for Top Tabs grouping.                  |
| Description Field       | Dropdown    | Column for document details/summary.           |
| Items per page          | Text        | Number of items to show per page (Pagination). |
| Pinned Column           | Dropdown    | Column determining pinned status (Yes/No).     |

### 3. Request Access Settings

Configures the feedback and access tracking workflow.

| Property                   | Type        | Description                                   |
|----------------------------|-------------|-----------------------------------------------|
| Show Request Access Column | Toggle      | On/Off switch for the entire request feature. |
| Select Request Site        | Site Picker | The site hosting the tracking list.           |
| Requests List              | Dropdown    | The list where requests are logged.           |

|                                  |           |                                           |
|----------------------------------|-----------|-------------------------------------------|
| <b>Email Field</b>               | Dropdown  | Field to store requester's email.         |
| <b>File ID Field</b>             | Dropdown  | Field to store the requested Document ID. |
| <b>Request ID Field</b>          | Dropdown  | Field to store the unique GUID.           |
| <b>Request Date Field</b>        | Dropdown  | Field to store the timestamp.             |
| <b>Reminder Field</b>            | Dropdown  | Field to track if a reminder was sent.    |
| <b>Already Requested Message</b> | Multiline | Toast message for duplicate requests.     |
| <b>Reminder Sent Message</b>     | Text      | Toast message for successful nudges.      |

## Permission Requirements

### Source Document Library

- **Access: Read-Only** for visitors.
- **Rationale:** Prevents modification of official documents.

### Request Access List

- **Access: Contribute.**
- **Item-Level Security:**
  - *Read access:* "Read items that were created by the user"
  - *Edit access:* "Create items and edit items that were created by the user"
  - **Rationale:** Privacy and integrity; users cannot see or modify others' requests.

## 8. Build & Deployment

### Build Commands

```
# Install dependencies
npm install

# Bundle for production
gulp bundle --ship

# Package solution (.sppkg)
gulp package-solution --ship
```

### Deployment

1. Upload .sppkg to **App Catalog**.
2. Deploy and trust the solution.
3. Add web part to page.

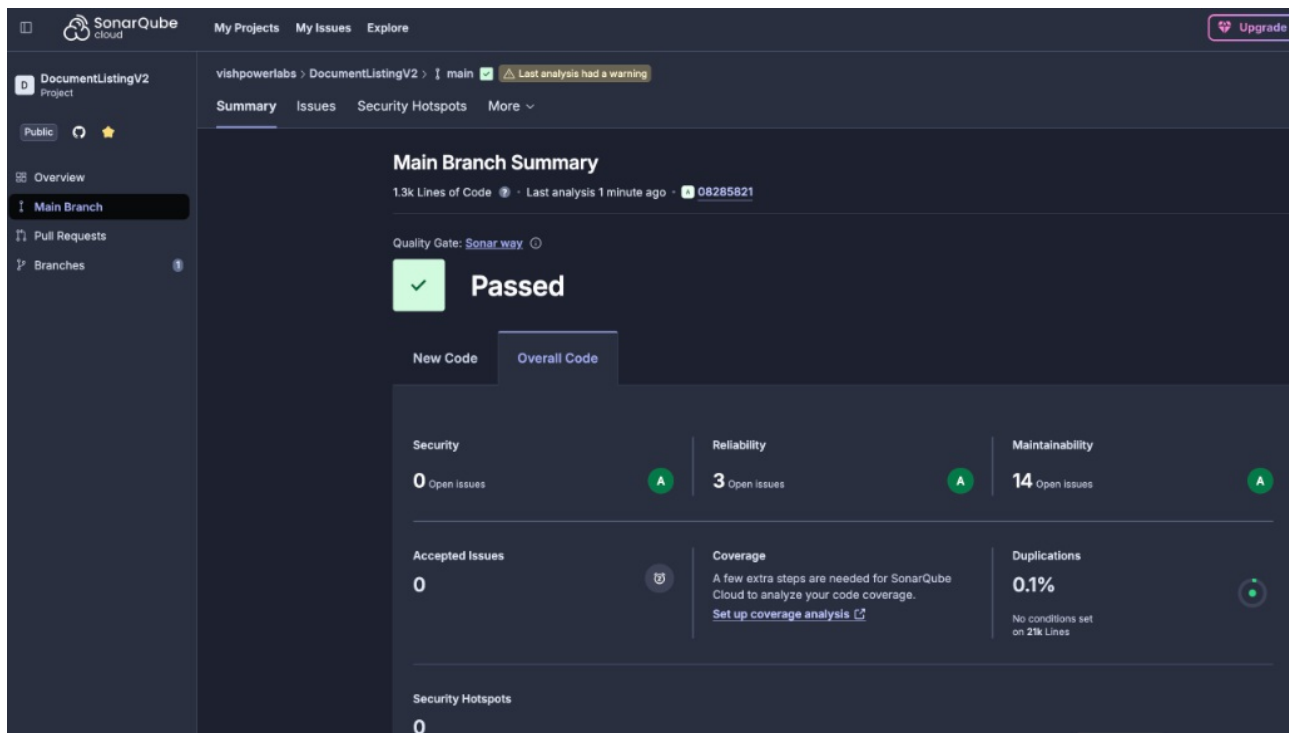
## 9. Future Roadmap

- **Feedback Collection:** integrated rating dialog in the web part.
- **Usage Analytics:** Dashboard for most requested documents.
- **Auto-Approval:** Immediate access for low-risk categories.

## 10. Code Quality & Security (SonarQube)

The solution is continuously monitored for code quality and security vulnerabilities using **SonarCloud/SonarQube**.

## Latest Scan Results



- **Quality Gate:** Passed
- **Security:** A Rating (0 Open Issues)
- **Reliability:** A Rating (3 Low-priority issues)
- **Maintainability:** A Rating (14 Code Smells - Minor)
- **Duplications:** 0.1% (Excellent)

This ensures the codebase remains secure, maintainable, and free of critical bugs before deployment.